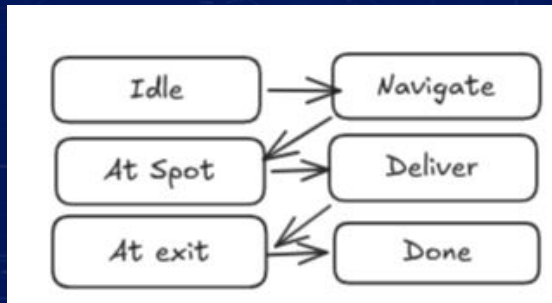
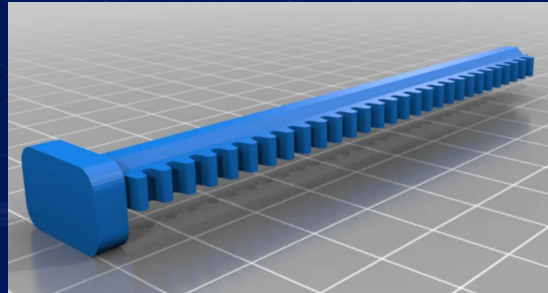


State Machine & PID Control



- 9-state machine sequences every mission phase automatically
- PID steers by correcting motor speeds based on marker offset from frame center
- Spins to search and re-acquires marker if lost mid-navigation
- ESTOP cuts all motors instantly from any state

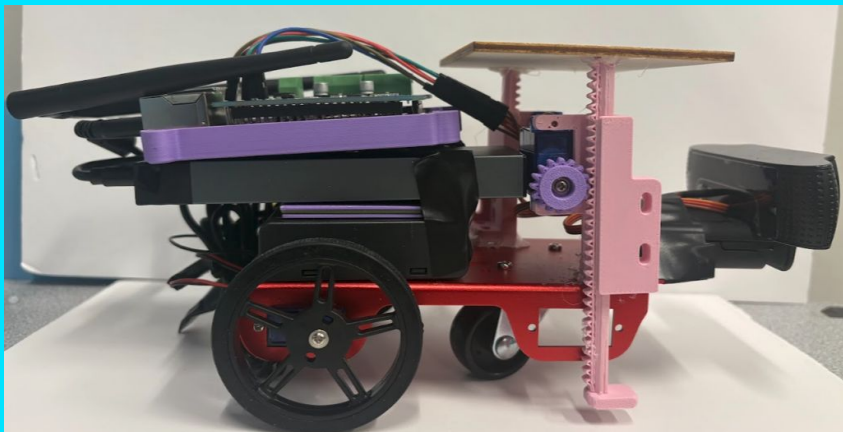
Hardware Design



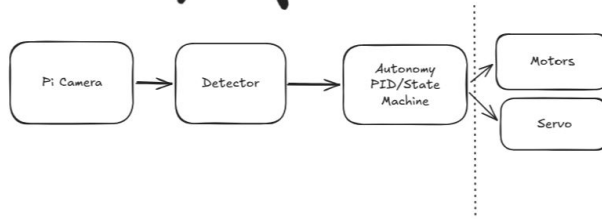
- Lift mechanism custom CAD-modeled and to precisely raise a vehicle payload
- Designed cases for the pi to attach to the frame
- Servo-driven actuation via PWM – no off-the-shelf lifting solution used



System Overview



Remote Control via ssh



System Architecture

- Fully autonomous: no human intervention required once mission is started
- Real-time visual navigation with no GPS, line-following tape, or external sensors
- Handles pickup, parking, retrieval, and return as a continuous mission
- Remotely monitored via live video stream and status dashboard (Done via ssh, tmux, and bash scripts)
- Custom CAD-designed and 3D-printed lift mechanism raises and lowers the vehicle with precision using servo-driven actuation

Autonomy

- Single button press – no further human input required
- Reacts to live camera feed every frame, not a fixed path
- Searches and re-acquires target marker if lost mid-navigation
- Lifts, repositions, delivers, and returns entirely on its own
- Operated wirelessly over SSH with live video stream



Computer Vision Pipeline

- Detects 4 HSV color markers: purple (exit), green (home), blue (park 1), red (park 2)
- Computes blob centroid offset from frame center steering in real time
- Stops automatically when marker area exceeds a tuned threshold
- Runs at 30 fps entirely on-device

